RAJALAKSHMI ENGINEERING COLLEGE

Jithisha G 251801119-2025-AIDS

J2

Started on	Thursday, 6 August 2026, 10:08 AM
State	Finished
Completed on	Thursday, 6 August 2026, 10:21 AM
Time taken	12 mins 44 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
{
    int i= 1;
```

```
    int s =1;
```

```
    while(s <= n)
    {
         i++;
         s += i;
     }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
int main(){
    int n;
    scanf("%d",&n);
    int c=0;
    int i=1;
    c++;
    int s=1;
    c++;
    while(1){
        c++;
        if(s<=n){
            i++;
            c++;
            s+=i;
            c++;
        }
        else{
            break;
        }
    }
    printf("%d",c);

}
```

	Input	Expected	Got	
✔	9	12	12	✔
✔	4	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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J2

Started on	Thursday, 6 August 2026, 10:28 AM
State	Finished
Completed on	Thursday, 6 August 2026, 10:46 AM
Time taken	18 mins 31 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
      printf("*");
    }
    else
    {
     for(int i=1; i<=n; i++)
     {
       for(int j=1; j<=n; j++)
       {
          printf("*");
          printf("*");
          break;
       }
     }
   }
 }
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
int main(){
    int n;
    scanf("%d",&n);
    int c=0;
        c++;
        if(n==1){
            c++; //printf("*");
        }
        else{
            for(int i=1;i<=n;i++){
                c++;//i<=n;
                for(int j=1;j<=n;j++){
                    c++; //j<=n;
                    c++; //printf("*");
                    c++; //printf("*");
                    break;
                }
                c++;
            }
            c++; //final outer-loop condition
        }
    printf("%d",c);
}
/*
#include<stdio.h>
int main(){
    int n;
    scanf("%d",&n);
    if(n==1){
        printf("1");
    }
    else{
        int c=5*n+2;
        printf("%d",c):
```

```
    }
}
*/
```

	Input	Expected	Got	
✔	2	12	12	✔
✔	1000	5002	5002	✔
✔	143	717	717	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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J2

Started on	Thursday, 6 August 2026, 10:47 AM
State	Finished
Completed on	Thursday, 6 August 2026, 10:52 AM
Time taken	5 mins 7 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
   for (i = 1; i <= num;++i)
   {
    if (num % i== 0)
       {
         printf("%d ", i);
       }
    }
 }
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
#include<stdio.h>
int main(){
    int n;
    scanf("%d",&n);
    int c=0;
    c++;
    for(int i=1;i<=n;++i){
        c++;
        if(n%i==0){
            c++; //printf("%d",i);
        }
        c++;
    }
    printf("%d",c);
}
```

	Input	Expected	Got	
✔	12	31	31	✔
✔	25	54	54	✔
✔	4	12	12	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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Jithisha G 251801119-2025-AIDS

J2

Started on	Thursday, 6 August 2026, 10:52 AM
State	Finished
Completed on	Thursday, 6 August 2026, 11:08 AM
Time taken	16 mins 37 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
#include<stdio.h>
int main(){
    int n;
    scanf("%d",&n);
    int c=0;
    c++;
    for(int i=n/2;i<n;i++){
        c++;
        for( int j=1;j<n;j=2*j){
            c++;
            for(int k=1;k<n;k*=2){
                c++;
                c++;
            }
            c++;
        }
        c++;
    }
    c++;
    printf("%d",c);
}
```

	Input	Expected	Got	
✔	4	30	30	✔
✔	10	212	212	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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J2

Started on	Thursday, 6 August 2026, 11:09 AM
State	Finished
Completed on	Thursday, 6 August 2026, 11:21 AM
Time taken	12 mins 11 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
   int rev = 0, remainder;
   while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n/= 10;

    }
print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
#include<stdio.h>
int main(){
    int n;
    scanf("%d",&n);
    int c=0;
    int rev=0;
    c++;
    int rem;
    c++;
    while(n!=0){
        c++;
        rem=n%10;
        c++;
        rev=rev*10+rem;
        c++;
        n/=10;
        c++;
    }
    c++;
    printf("%d",c);
}
```

	Input	Expected	Got	
✔	12	11	11	✔
✔	1234	19	19	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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